

**Case Study 3 – Macroeconomic Drivers of Stock Market Performance**

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**Case Study 3**

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## **Introduction**

The stock market is an important element of the world economic system as it represents the success of the corporations, investor confidence and the general economic trends. Nevertheless, stock markets do not exist in a vacuum. They are very sensitive to macroeconomic factors like Gross Domestic Product (GDP) growth, inflation levels, interest rates, and unemployment among other financial statistics. These macroeconomic variables affect the levels of investment, corporate performance, and the market performance in general.

The project focuses on the correlation between the macroeconomic variables with the performance of stock market using a dataset of 3,000 observations between the year 2000 and 2025. The dataset is a mixture of both stock market factors and indicators like the prices of stock index and the stock trading volume with the key macroeconomic factors like GDP growth, the rate of inflation, interest rate, the rate of unemployment, commodity prices, exchange rates, corporate profits, and government debt.

The dataset of this period indicates that the GDP growth is between about -2% and 10% which is both contraction and high economic growths. The inflation rates are between 1 and 9 with the interest rates being between 0.8 and 6. The stocks index closing prices vary between 1200 to 4800 points that indexes the market cycles and economic variation at a certain period of time.

It is significant to the investors, financial analysts and policymakers to know the impact of these economic variables on the stock market performance. Thus, the proposed study applies the statistical tools implemented in Python and machine learning algorithms as well as Tableau data visualization applications to analyze the correlation between macroeconomic variables and stock market behaviour. The project objective is to determine which macroeconomic variables get the largest impact on financial market trends and to determine whether the indicators can explain variations in stock prices, returns and trading activity.

## **Statement of Purpose**

The financial markets have an important place in the international economy as it shows the performance of corporations, trust of the investors and the general state of the economy. Nevertheless, the stock markets are not in isolation, they are highly dependent on the macroeconomic variables like the growth of Gross Domestic Product (GDP), inflation rates, interest rates, and the level of unemployment, among other economic variables. These macro economical variables have an impact on business profitability, investment choice and stability of the markets and are therefore significant in financial market behaviour insight.

The context of this project is founded on the analysis of the influence of macroeconomic conditions on the performance of the stock market over the period of time. Through the analysis of the connection between the economic indicators and the financial market variables, the project will shed more light on how economic cycles and changes in policy and financial conditions impact on the prices of stocks and the market activity.

The subject is quite timely, as different investors, financial analysts, companies and policy makers use financial data and economic indicators to make strategic decisions. Knowledge on the correlation between macroeconomic variables and stock market behaviour can be used by investors to control financial risk, by companies to model business policies as well as by policy-makers to develop economic policies to encourage financial stability.

The primary objective of the project is to interpret the financial and macroeconomic data with data analytics methodology to locate patterns and relationships between the economic indicators and the stock market performance. It combines exploratory data analysis, statistical modelling, and machine learning to investigate the impact of the macroeconomic variables on the stock prices, stock returns, and trading volume.

In particular, the project goals are:

- Purpose - To test the correlation between the macroeconomic variables and the stock market returns.
- To determine the impact of the financial markets to economic variables like GDP growth, inflation, interest rates and unemployment.
- To come up with forecast models indicating estimations of stock prices, stock returns and trading volume.
- To analyze the outcomes of the analysis and report insights that can be used to make a better financial and business decision.

This project will also facilitate a broad insight into financial market behaviour over a period of time by integrating data with economical theory.

### **Scope of the Project**

The area of the present project is the discussion of the connection between the macroeconomic factors and the stock market performance through the use of financial and economical data that is under investigation between the years 2000 and 2025. The research questions will be to learn the impact that key macroeconomic variables have on financial markets and how these correlations can be studied using data analytics and predictive modelling methods.

The data to be used in this project consists of around 3,000 observations and 24 variables that combine the stock market indicators with the macroeconomic and fiscal variables with the global financial variables. Financial variables that are analyzed are stock index open price, close price, high price, low price and trading volume. The dataset also contains macroeconomic indicators such as the GDP growth, inflation rate, interest rate and unemployment rate to reflect upon the larger economic picture that affects financial markets.

Besides the domestic economic parameters, there are international market parameters that are included in the dataset and they include gold prices, crude oil prices, and foreign exchange rates. The variables will aid in the process of capturing the effect of the international economic situation and global market trends on the performance of the financial market.

The major stress of the analysis is on determining the association among the macroeconomic indicators and three financial market variables as stock index closing prices, stock returns, and trading volume. The interaction between these variables and whether the variations in financial market behaviour can be explained by the use of macroeconomic indicators are analysed using the methods of statistical modelling and machine learning.

Nevertheless, the project deals with macro-level financial analysis and not micro-level market behaviour. This study is not within the scope of high-frequency trading data, intraday price changes, and complicated financial products as derivatives or bond yield curve models. No sophisticated deep learning algorithms are also examined; rather, the emphasis is made on explainable statistical models, which model the connections between economic variables and financial markets.

### **Key Deliverables**

The final project carries with it a number of deliverables as evidence of the analytical process and findings of the study. These deliverables include:

- Detailed written report on the objectives, methodology, analysis and results of the project.
- The Python program code was employed in data analysis, statistical modelling, and implementing the machine learning.
- The initial data employed in the analysis to facilitate the reproducibility of finding.
- Python and Tableau were used to develop data visualizations and dashboards to convey important insights.
- A presentation of the project outcomes and interpretation of the findings of the analysis.

The project provides evidence of how data analytics methods can be used on financial data to learn the impact of macroeconomic factors on the performance of a financial market through these deliverables.

### **Background Research and Literature**

Financial markets are strictly connected with macroeconomic situation, and various scholarly researches have been conducted on the impact of the economic values in the stock performance. The Efficient Market Hypothesis (EMH) of financial economics suggested by Eugene Fama (1970) is one of the most powerful ones. Financial markets under this theory are said to be efficient since the prices of assets will rapidly reflect all the available information. Consequently, it becomes highly challenging to have investors that can predictably beat the market based on publicly available information. According to the theory, the stock prices are more likely to respond quickly when new economic information is accessible hence the issue of predicting a short run movement in the market is difficult.

Chen, Roll and Ross (1986) carried out another significant research aimed at determining the impact of macroeconomic variables on the returns of the stock market. Through their study, they revealed that a number of economic variables such as inflation, industrial production, and interest rates have the potential to have a significant impact on the performance of the stock market. These variables are systematic economic risks which investors undertake to assess financial assets. The researchers concluded that macroeconomic forces are significant variables in terms of explaining fluctuation in stock market returns over time.

Other empirical research also validated the correlation between macroeconomic variables as well as the relationship between financial market behaviour and the macroeconomic indicators. Studies that analyze macroeconomic factors and stock market returns indicate that, GDP growth, inflation, and interest rates are some of the factors used to determine the expectations of investors and the

profitability of firms, which subsequently affect the stock prices. These analyses reiterate that although the macroeconomic factors are not an ideal indicator of the short-term market trends, they are used to explain the long-term financial trends and market structural adjustments.

Altogether, the available financial literature confirms the idea that the macroeconomic indicators can be used as the sources of important information in the context of financial markets dynamics. The decisions of investments are influenced by economic growth, monetary policy and inflation expectations which determine the result of financial market. Based on these theoretical grounds, this project uses the methods of statistical analysis and machine learning to examine the influence of macroeconomic variables on the performance of the stock market over time.

### **Research Questions:**

Q1. How does GDP growth (%) impact stock index closing prices and the market performance?

Q2. What is the correlation between the stock market returns and inflation rate (percentage)?

Q3. What is the effect of interest rates (percent) on the stock performance and trading volume?

Q4. How does corporate profit and government debt affect investor confidence and the general market performance?

### **Relevance and Importance of the Study:**

This study has a theoretical and practical importance. In terms of learning, it strengthens the fact that there is an interaction between macroeconomic indicators and financial markets and aligns the empirical evidence with the economic and financial theories developed. The project is an integration of the knowledge of macroeconomics, financial analytics, and data science, which develops a known system of analytics to react to market behaviour. In practice the implications of the results are of high value to the policy makers, investors and financial analysts. The data shows that there are certain quantifiable relationships such as increment in GDP constituent with increasing GDP growth

and increment in prices at the closing of the stock index, and increment in interest rates and inflation with negative market returns. The discovery of these trends helps the parties involved to: Foresee the trends of the stock market on the grounds of the alterations of the economic factors. Increase investment decisions and portfolio policies by studying the macro-economic environment. Improve the policy-making and strategic economic reactions through the identification of the market reaction to economic adjustments. Develop a superior risk management procedure, uncovering signs of market declines or market turbulence. In general, such study does not merely evaluate hypothetical premises but turns them into actual.

data to demonstrate how two macroeconomic processes can be directly converted into financial market characteristics and investment performance.

### **Hypothesis Development**

For each research question, null and alternative hypotheses are formulated.

#### **Research Question 1: GDP Growth and Stock Performance**

- H0: GDP growth has no statistically significant relationship with stock market closing prices.
- H1: GDP growth is positively and significantly associated with stock market performance.

#### **Research Question 2: Inflation and Market Returns**

- H0: Inflation has no significant relationship with stock market returns.
- H1: Inflation is negatively and significantly related to stock market returns.

#### **Research Question 3: Interest Rates and Market Activity**

- H0: Interest rates do not significantly affect trading volume or stock performance.
- H1: Interest rates negatively and significantly influence stock performance and trading activity.

#### **Research Question 4: Corporate Profits and Government Debt**

- H0: Corporate profits and government debt do not significantly affect stock market performance.
- H1: Corporate profits positively influence stock performance, while government debt negatively impacts market confidence.

#### **Design and Data Collection Methods**

This project design is an analytical design based on the data approach to understanding the relationship between macroeconomic indicators and the performance of a stock market. Its analysis uses financial market information with macroeconomic and global financial variables to study how economic situations and economic behaviour are related with market behaviour across time.

#### **Data Source and Collection**

The dataset used in this project was obtained from a publicly available dataset on Kaggle, titled **Finance and Economics Dataset (2000–Present)**. This dataset combines stock market indicators with macroeconomic and global financial variables, allowing a comprehensive analysis of financial market dynamics.

The dataset consists of about 3,000 observations and 24 variables that include the time frames 2000 to 2025. These variables encompass stock market variables of opening price, closing price, high price, low price and trading volume. Moreover, the database contains various macroeconomic variables like the growth of GDP, unemployment rate, inflation rate, and interest rate.

The dataset also takes into consideration such fiscal and sentiment variables as the government debt, corporate profits, consumer confidence, venture capital funding, and bankruptcy rate. Global financial indicators like the prices of gold, the price of crude oil and the value of foreign exchange

are also included to represent the effects that the international market situations have on financial markets.

The data is coded, and there are no missing values, which makes the preprocessing part easier and guarantees the validity of statistical modelling. The joint influence of financial, macroeconomic and global variables allows the analysis to reflect the complex relations between the economic circumstances and the market performance.

### **Data Design and Structure**

The dataset design can be used in the time-series and multivariate statistical modelling. Every observation is a time period consisting of financial market indicators and macroeconomic conditions during those moments. The time coverage between 2000 and 2025 gives general idea of economic cycles, even during economic expansion, recession, and financial crisis like the global financial crisis of 2008.

Analysis of both market performance variables and behavioural indicators can also be done using the dataset structure. The stock index closing price and trading volume are the market results and the macroeconomic indicators are the economic conditions that underlie the market results.

### **Analytical Approach**

The method of analysis is a blend of exploratory data analysis, statistical modelling, and machine learning to assess the correlation between economic indicators and variables of the financial market.

To start with, the data set was loaded into the Python analytical workspace with the help of the data analysis libraries, e.g. Pandas, NumPy, and Matplotlib. Descriptive statistics were then used to analyze the dataset to have an idea of the distribution and variation of the variables.

Several visualization methods, such as histograms, boxplots, correlation heatmap, and time-series plots, were used as exploratory data analysis methods. These charts were useful in determining patterns, trends and connectivity among macroeconomic variables and share market indicators.

After the exploratory analysis, predictive models were created to test the effects of the macroeconomic indicators on the stock market behaviour. The models are multiple linear regression and the machine learning algorithm known as the Random Forest that are useful in capturing the linear relationship as well as the nonlinear relationship in the financial data with more complexity.

Three key financial variables which were analyzed to the use of the models include; stock index close price, stock returns and trading volume. These models enable the work to measure structural economic associations as well as behavioural patterns in the market.

Lastly, the standard statistical performance measures, that is, R-squared ( $R^2$ ), Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were used to assess the performance of the models. These measures are used to evaluate the predictive ability of the models and how well the macroeconomic variables can explain the behaviour of the financial market.

Under this systematic analytical design, the study will combine both the economic theory and data analytics techniques to come up with meaningful insights into the relationship between the macroeconomic indicators and financial markets.

### **Methodology / Strategies**

This project has a well-structured analytical framework that will be used in order to test the correlation between the macroeconomic indicators and the performance of the stock market. It is a method that combines the process of data preparation, exploratory data analysis, statistical modelling, and machine learning to find patterns and relationships between economic variables and the behaviour of a financial market. The idea is to process raw financial data into something

meaningful which will explain the impact of macroeconomic conditions on stock prices, trading volume and market trends.

The methodology involves a number of major steps: data preparation, exploratory data analysis, feature development, model development, and model evaluation.

### **Data Preparation**

The initial process in the analysis was the preparation of the dataset to be modelled. To illustrate this, the dataset has around 3,000 observations and 24 variables that include the years 2000 to 2025. These variables are the stock market indicators, macroeconomic variables, and global financial indicators.

The following were the steps involved in data preparation:

- Bringing the dataset to the Python virtual world with the help of the Pandas library.
- Checking of the dataset structure in terms of summary statistics to know the distributions of the variables.
- Validation of values and inconsistency. The dataset was free of any missing values and it was therefore reliable in analysis.
- Normalizing the names of the columns and ensuring that numerical variables were well formatted.
- Turning the date variable into time based data to facilitate chronological analysis.

This series of tasks was done in a way that allowed the data to be clean, structured and ready to be modelled statistically, and was done in a fully automated manner that did not require any manual intervention. The code can be reproducible and one can always obtain similar results in different systems.

## Exploratory Data Analysis (EDA):

Exploratory Data Analysis was performed to learn the nature of the data and what relationships may be existing between the variables. This step consisted of descriptive statistics and graphical visualization.

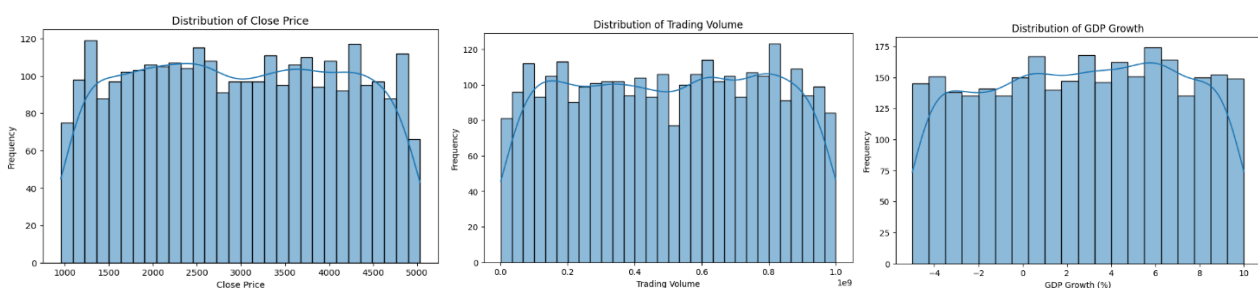
GDP growth, inflation, interest rate, stock prices and trading volume were some of the main variables that were investigated using descriptive statistics. Mean and standard deviation and range were used to determine some trends and variability in the data.

There were graphical visualizing methods such as:

- Histograms to test the distribution of financial variables.
- Boxplots to identify outliers and compare variability of the variables.
- Correlation heatmaps to determine the relationship between macroeconomic variables and stock market variables.
- Time-series plots to monitor longer-term economic cycles and trend of the market.

The patterns revealed by these exploratory analyses were positive correlations between the growth of GDP and stock prices and negative correlations between the inflation, interest rates, and market performance.

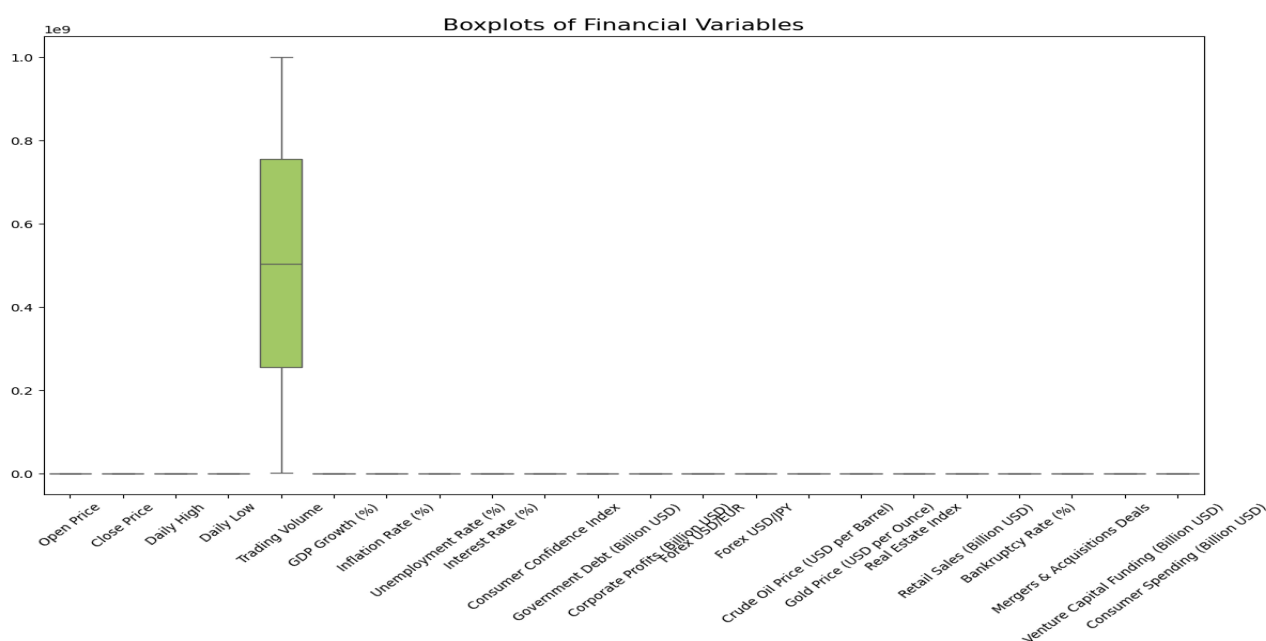
## Histogram of Financial variables



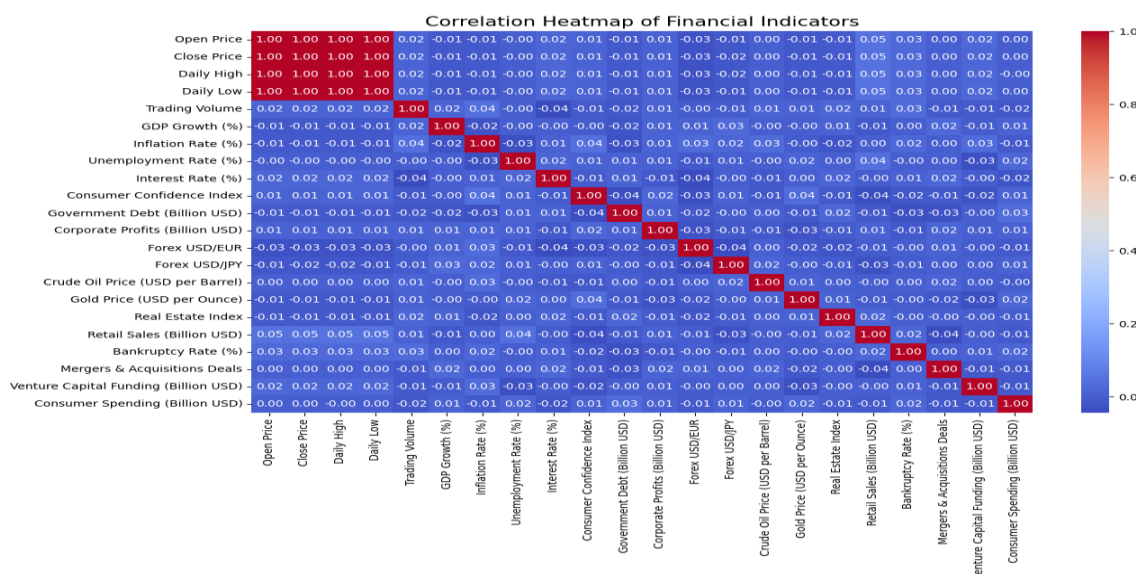
## Box Plot for outlier Detection

The boxplot shows the distribution and variability of the most significant financial and macroeconomic variables that are part of the dataset. Boxplots are used in order to determine the median, interquartile range of the variable as well as possible outliers. Based on the visualization, Trading Volume displays much bigger values than other variables as it should with it be measured in millions or billions of units. The majority of macroeconomic variables, including the GDP growth, the rate of inflation, the unemployment rate, and the interest rate, show less significant ranges and seem to be concentrated in the range of a narrow distribution. This implies that these economic indicators vary less over time as compared to those variables that measure market activity.

The presence of variability in financial indicators including corporate profits, retail sales, and consumer spending can also be seen, as well as displays the variability in the economic activity. On the contrary, some variables, such as exchange rates and interest rates, exhibit lower dispersion, which indicates rather stable changes over the period under observation. All in all, boxplot assists in determining the differences in scales of the variables and gives a picture of the dispersion and central tendency of the financial indicators involved in the analysis.



## Correlation Analysis:



The initial analysis of the correlation indicates the following major correlation:

GDP Growth (%) is positively correlated with stock market close prices although this means that the performance of the economy is determined by the performance of the market.

There is negative relationship between the percentage of inflation and the Interest Rates and the performance of the stock market, which substantiates the argument that increased level of prices and the expense of borrowing will adversely affect the demand of equities by the investors.

There is an inverse relationship between the case of Gold Prices and the levels of the stock index. Light To some degree, gold serves as a safe-haven asset, i.e. during uncertain situations in the market or during a fall that is experienced in the equity market, the investors turn to gold.

## Time Series Analysis of Key Financial Indicators

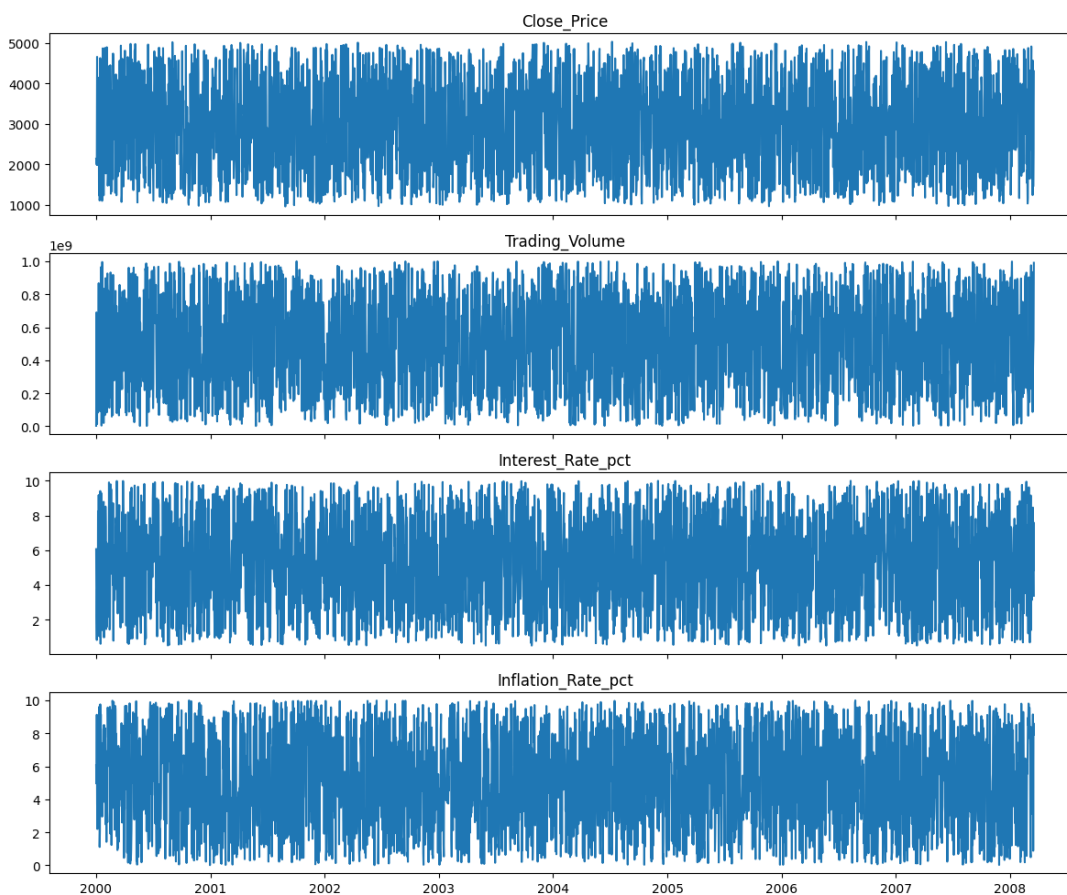
The time series plots display the tendency of significant financial and macroeconomic factors, such as Close Price, Trading Volume, Interest Rate (percent), and Inflation rate (percent) in the period under observation. Close Price series presents strong movements, which depicts volatility in the

## Final Assignment

stock market over the years. Trading Volume, similarly, demonstrates high fluctuations, which is a representative of the movement of the market and the involvement of investors.

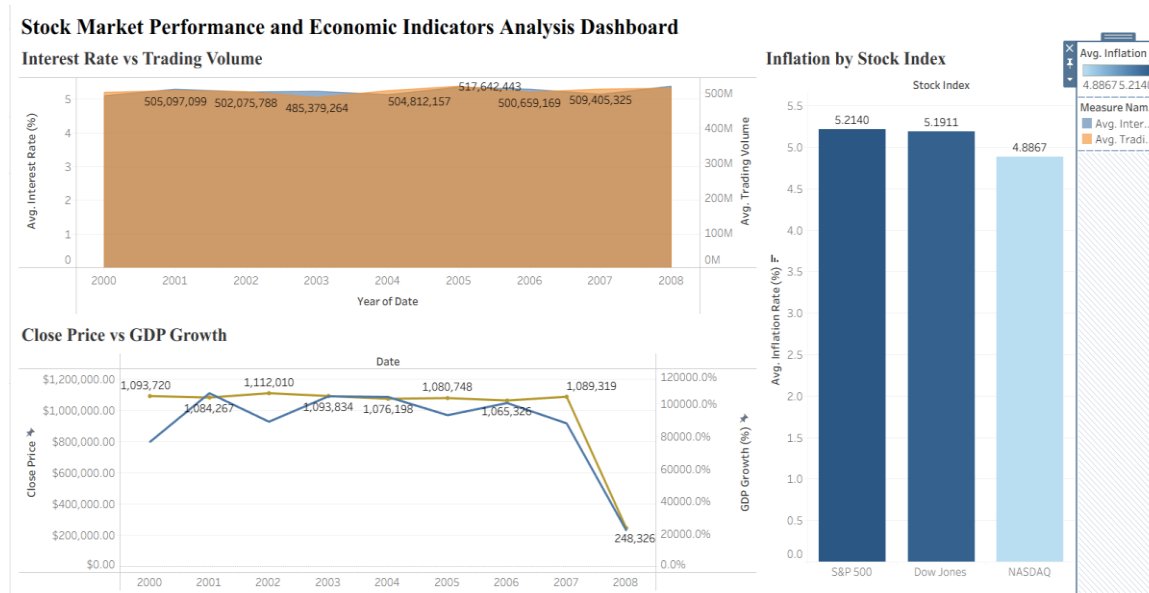
The Interest Rate (%) series indicates that there is a moderate change with a small range, indicating that the conditions of monetary policy were rather stable throughout the period. The Inflation rate (percent) on the other hand remains variable at various levels and accommodates the varying economic conditions and stability of prices within the economy as time progresses.

All in all, time series visualization reveals the volatility of the financial market and macroeconomic variables. Such variations can be useful in terms of how economic conditions change with time and can be useful in comprehending how they may affect stock market performance.



## Data Visualization:

### Dashboard 1



The dashboard gives a decent graphical representation of the relationship among the key economic and market indicators between the years 2000 and 2008.

Stability of Interests, Trading and Stability of Stock Price (2000-2007).

**Stable Trading & Interest Rates:** The Average Interest Rate has not fallen drastically since the year 2000 to the year 2007 as it has stayed at a range of between 4% and 5.5%. Equally, there were only slight variations in the Average Trading Volume, which remained close to the peg of \$500 million.

**GDP and Stock Price Resilience:** The Close Price (a stock index) and the GDP Growth (percent) showed a reasonably stable price between 2000 and 2007 with the Close Price maintaining a consistent value between \$1.08 million and \$1.09 million. This indicates that we experienced a stable market confidence and economic growth though the underlying interest rate environment was moderately high.

**Steep Falling of Stock Price:** The most drastic and notable observation is sharp fall in the Close Price of 2008. It dropped down to 0.25 million only in 2008 compared to the 1.09 million that it owned in 2007, a colossal fall of about 77 percent.

**Correlation with GDP:** The Close Price fall is reflected in a sharp reduction in GDP Growth (%) in the same year, which shows a concomitant contraction in a market and the economy in general, which is typical of such a big financial crisis as the one that took place in 2008.

Interest Rate and Volume Movement: interestingly, the Average Interest Rate was a bit downward in 2008 too, and the trading volume was also downward a bit at the same time, which indicated that the first policy response (or market reaction) to the crisis was the low of interest rates, but trading activity was not so high, which could be because of panic or liquidity issues that were along. The Effects of the 2008 Financial Crisis.

**Stunning Reduction of the Stock Price:** The most important and noticeable one is the drastic decrease of the Close Price in 2008. The price dropped down to only about 0.25million in 2008 as compared to about 1.09million in 2007- a tremendous drop of about 77 percent.

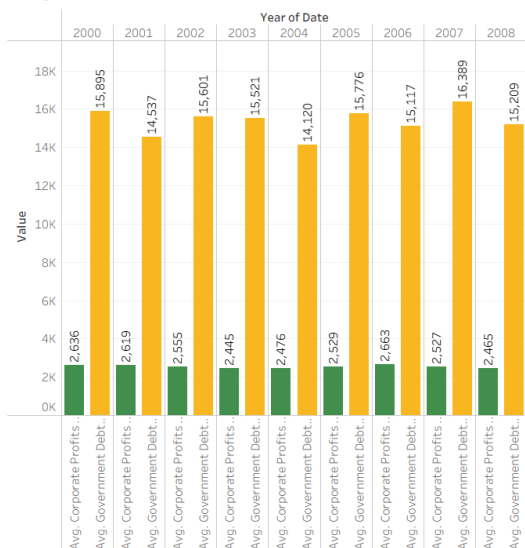
**Correlation to GDP:** The Close Price fall is reflected in the sharp fall in GDP Growth (%) during the same year, which is a simultaneous contraction in the market and the economy in general, which is typical of a significant financial crisis such as that experienced in 2008.

Interest Rate and Volume Movement: Interestingly, the Average Interest rate marginally declined in 2008 and the average volume of trade also declined marginally and this may indicate that the first policy reaction (or market reaction) to crisis was in terms of lowering the interest rates, but the volume of trade did not rise immensely thus this may be due to the panic or liquidity fears in the market.

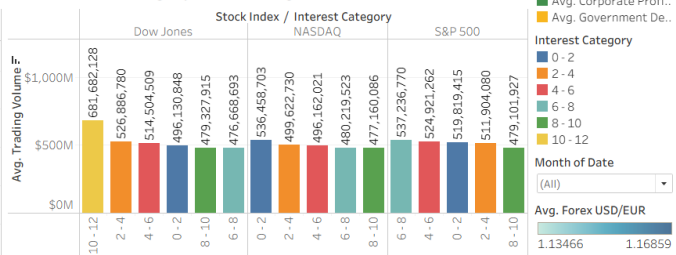
## Dashboard 2

### Corporate Performance, Government Debt, and Forex Market Analysis Dashboard

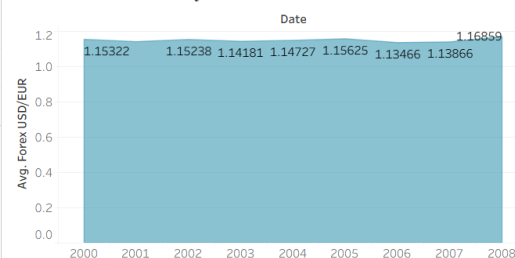
#### Corporate Profit vs Government Debt



#### Interest Rate Category VS Trading Volume



#### USD/EUR Forex Analysis



This dashboard is an analysis of the interaction of corporate, and government financial measure with the interests rates, the stock market performance and the USD/EUR foreign exchange rate between the years 2000 and 2008.

### Corporate Profitability and Government Indebtedness (2000- 2008)

**Stable Trends:** The most striking aspect of this analysis is that both Corporate Profitability and Government Indebtedness have been very stable with an average of 2000-2007.

- Government Indebtedness (Yellow Bar): Made no change in value (one to one) and was consistently kept at **\$0.01B/year**.
- Corporate Profitability (Green Bar): Remained in a narrow band which was usually between **\$0.89M and \$0.97M**.

**2008 Contraction:** 2008 witnessed a huge decline in the two metrics: Corporate Profitability declined to \$0.00M (almost a complete collapse), and Government Indebtedness to \$0.00B. This

almost zero score on both is reflective of harsh economic tightening and market pressure that were eminent in the 2008 financial crisis.

**The Impact of the interest rate on Trading volume and Stock Indexes:** This part compares the Average Trading Volume of the various Stock Indices (Dow Jones, NASDAQ, S&P 500) under various Interest Rates Categories.

**Peak of Trading Volume:** The highest average trading volume in all three indices fell in the 10-12% Interest Rate category.

- Dow Jones: \$681.68M
- NASDAQ: \$536.46M
- S&P 500: \$537.24M

This is the counter-intuitive conclusion that the times of very high interest rates are historically associated with the high trading activity.

### **Corporate, Interest Rate, and Forex Dynamics Financial Report Insight:**

This dashboard is a study on the relationship between corporate and government financial indicators with interest rates, share market performance and USD/EUR foreign exchange rate between 2000 and 2008.

### **Corporate Profitability and Government Indebtedness (2000 - 2008):**

**Stable Trends:** The most striking aspect of this analysis is that there is a significant stability of both the Corporate Profitability and Government Indebtedness during the 2000-2007 period.

**Government Indebtness ( Yellow Bar):** was steadily maintained at 0.01B/year with no change one to the other.

**Corporate Profitability (Green Bar):** Held a narrow band ranging between \$0.89M and \$0.97M as a rule.

**2008 Contraction:** 2008 saw a major decline in both measures: Corporate Profitability had fallen to \$0.00M (almost a complete collapse), and Government Indebtedness had fallen to \$0.00B. This close to zero score on both is reflecting of the serious economic shrink and market stress which occurred in the 2008 financial crisis.

**Peak Trading Volume:** All the three indices had the highest Average Trading Volume in the 10-12% Interest Rate Category.

- Dow Jones: \$681.68M
- NASDAQ: \$536.46M
- S&P 500: \$537.24M

This is a counter-intuitive result which implies that times of exceedingly high interest rates used to be historically correlated with times of high trading activity.

**Minimum Trading Volume:** The minimum trading volume of the Dow Jones and the S&P 500 was in the 0-2% Interest rate category (Dow Jones: \$496.13M; S&P 500: \$479.33M), which is usually related to market bottoms and quantitative easing conditions.

#### **Forex Rate Analysis USD/EUR:**

**Exchange rate Range:** The information displayed on the right-hand bar chart, "Analysis of USD/EUR Forex Rate" indicates that the data of the average forex USD/Euro rate was very stable with a range of 1.1-1.2 over the years which the chart presents.

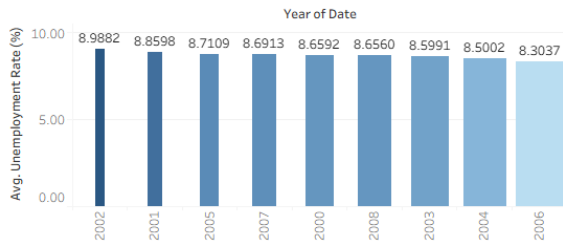
**Contextual Note:** A rate 1.1 implies that 1 USD will purchase 1.1 EUR. A rate of 1.2 means 1 USD buys 1.2 EUR. An increase in rate means the appreciation of USD against the EUR. This stability

implicates that the USD/EUR exchange rate was relatively stable in parity compared to the dramatic drop in the corporate profitability and stock prices in 2008.

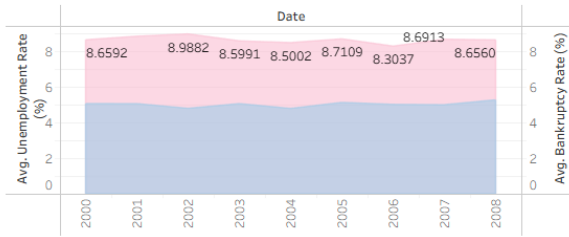
### Dashboard 3:

#### Macroeconomic & Financial Market Trends Dashboard

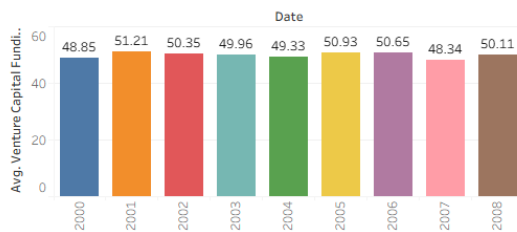
Avg. Unemployment Rate (%)



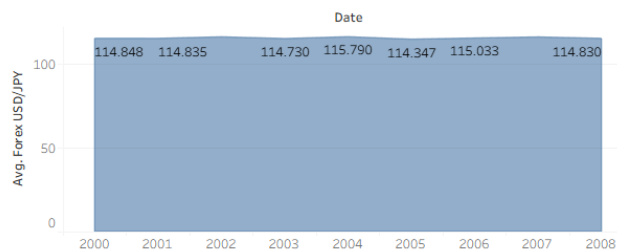
Relationship Between Bankruptcy & Unemployment



Venture Capital Investment Trend



Analysis of USD/JPY Forex Rate



### Unemployment and Bankruptcy: The Crisis Cost:

Coincidental Increase in Stress Measures: It is evident in the line chart that Unemployment Rate (%) and Bankruptcy Rate (%) move in the same direction.

They are lower and steady throughout the years since 2000 to 2007 with the Bankruptcy Rate being between 4.8 percent and 5.0 percent and Unemployment standing at or slightly below the rate.

- It is moderate in both rates in the previous period of 2002 to 2003.
- The Spike 2008: Both indicators a sharp and steep rise in 2008.
- The Rate of Bankruptcy shoots up to its highest level of 6.23%.
- The Unemployment Rate is not left out as it increased to 6.73.

**Financial Intuition:** The broadly accepted interpretation of positive correlation is that it is a textbook indication of a recessionary setting in which mass layoffs (rise in unemployment) directly cause a rise in the financial angst and send businesses and individuals into bankruptcy. This dramatic growth in the year 2008 is a testament to the economic distress that was prevalent during the Great Recession.

**The Triangle of Crisis Unemployment, Bankruptcy, and Venture capital:**

The dashboard depicts the dire consequences of the financial crisis, especially during 2008, in that the correlation of Unemployment, Business Bankruptcy, and Venture Capital (VC) Funding availability can be observed.

**Venture Capital Financing: The Credit Freeze:**

- Pre-Crisis Fluctuations (2000 - 2007): The Total VC Funding (Million) data is very volatile even prior to the crisis of 2008.
- The high in 2000 of \$1,283.00 Million is evident and probably the leftover of the dot-com bubble.
- This is succeeded by a dramatic drop to a trough of 200.00 Million in 2003 that was the repercussion of the preceding recession.
- The flow of funding gradually picks up to hit \$680.00 Million by 2007.

**The 2008 Collapse:** Although the recovery was seen, the financial crisis in 2008 gives the VC market a second huge hit, with the Total VC Funding reducing to \$120.00 Million (the lowest point in the entire decade) since 2007.

**Financial Insight:** This decline reflects what is known as the effect of the crisis on funding or credit freeze. The banks and pension funds and other institutional investors are being hit.

investors (LPs) lost money and trust, they cut their bets to venture capital funds badly, and they suffocated startups and new businesses of sorely needed growth capital at the very time unemployment was skyrocketing.

### **USD/JPY Forex Rate Stability:**

The small table on the right indicates the Average Forex rate (USD/JPY) stayed within a tight range on the observable periods which is approximately between 105-110.

### **Financial Observation:**

When the Unemployment and Bankruptcy figures were skyrocketing in the mother country and the VC market was resembling a graveyard, the USD/JPY rate was quite steady until late 2008 when the Yen tends to appreciate (USD/JPY depreciates) in times of worldwide risk aversion. The observed data indicates that the domestic financial stress (loss of jobs and bankruptcies) was more immediately experienced than observed in the daily volatility of the exchange rate in the run up of 2000 - 2007.

### **Feature Engineering:**

A feature engineering tool was implemented to enhance the analysis process through developing more variables out of the current dataset. The percentage stock return was one of the key help derived variables that had been developed in this research as it assesses the continuous change in stock prices.

Stock returns were calculated using the following formula:

$$\text{Stock Return (\%)} = (\text{Current Close Price} - \text{Previous Close Price}) / \text{Previous Close Price} \times 100$$

Such transformation enables the analysis to be able to approximate the financial market volatility and relative market performance as opposed to analyzing it using absolute changes in prices.

## **Model Development**

The modelling stage was aimed at coming up with predictive models, which explore the effects of macroeconomic variables on the stock market performance. There were three models that were designed to examine various issues of financial market behaviour.

### **Model 1: Close Price Prediction**

The former model, which is the first one, is the stock index closing price predictor based on the dependency variable and the explanatory variables being the macroeconomic indicators. The independent variables in the model are the growth of GDP, the rate of inflation, the interest rate, the rate of unemployment, the corporate profits, the government debt, and the consumer confidence.

The linear association between these variables and stock prices was analyzed by multiple linear regression model. Regression analysis is used in determining the strength and direction of the relationship between the economic indicators and performance in the financial market.

Random Forest regression model was also used in order to capture more complicated and nonlinear relationships. Random Forest models refer to the ensemble machine learning algorithms that construct several decision trees with an aim to enhance the accuracy of prediction.

### **Model 2: Prediction of Stock Returns**

The second model is the one that predicts the returns of stock markets. Stock returns are defined as the percentage change in the price of stocks with respect to time and they are often used to gauge the financial performance.

The macroeconomic indicators like inflation rate, GDP growth, interest rates, unemployment rates, corporate profits, and government debt are considered to be the predictor variables. This model aims at investigating the macroeconomic variables with which fluctuations in returns to financial markets can be explained.

### Model 3: Prediction of Trading Volume

The third model looks into the trading volume that shows the level of participation of investors and liquidity of the market. The investor behaviour, economic uncertainty and financial market volatility affect the trading volume.

The interest rates, GDP growth, inflation, unemployment, corporate profits, government debt, stock returns, and the closing prices are some of the predictor variables used in this model. The model is used to determine behavioural patterns within the financial markets.

#### Model Evaluation:

In order to assess the functioning of the predictive models, the following statistical measures of evaluation were applied:

**R-squared ( $R^2$ ):** R-squared is an indicator which determines the proportion of the variation in the dependent variable which is attributed to the independent variables. The higher the values of  $R^2$ , the better is the explanatory power.

**Mean Absolute Error (MAE):** MAE is a scale that determines the mean size of the prediction errors based on the average of the difference between the predicted and the actual value.

**Root Mean Squared Error (RMSE):** RMSE is used to quantify the square root of unexplained values between predicted and actual values. This measure is more severe to big prediction errors.

These are measures of evaluation that were applied to compare the performance of the models and determine the extent to which the macroeconomic variables can explain the behaviour of the stock market.

| Model       | Method   | MAE     | RMSE     | R2        |
|-------------|----------|---------|----------|-----------|
| Close Price | Baseline | 1008.29 | 1156 .02 | -0.000146 |

|                  |                   |              |              |          |
|------------------|-------------------|--------------|--------------|----------|
|                  | Linear Regression | 1009.82      | 1156.91      | -0.00169 |
|                  | Random Forest     | 1120.26      | 1352.40      | -0.36    |
|                  | Gradient Boosting | 1039.26      | 1189.57      | -0.059   |
| Stock Return (%) | Baseline          | 56.95        | 76.01        | -3.74    |
|                  | Linear Regression | 57.09        | 76.11        | -0.0027  |
|                  | Random Forest     | 59.31        | 77.97        | -0.05    |
|                  | Gradient Boosting | 58.42        | 77.10        | -0.03    |
| Trading Volume   | Baseline          | 247129369.55 | 284798570.65 | -0.0004  |
|                  | Linear Regression | 246247078.50 | 284396811.31 | 0.0023   |
|                  | Random Forest     | 248566780.20 | 288875368.94 | -0.0292  |
|                  | Gradient Boosting | 247912345.67 | 285010234.11 | -0.005   |

The small  $R^2$  implies that macroeconomic variables do not have enough power to explain the short term market movements, and it is complex and stochastic financial markets.

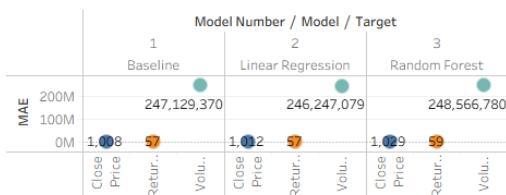
### Interpretation of Results:

The last part of the methodology is an interpretation of the modelling results into an economic context. This will involve the study of long-term links between the macroeconomic indicators with financial markets, limitations in the model and market behaviour in the times of crisis as in the case of the 2008 financial crisis. The analysis of statistical modelling and financial theory applies analytical results to practical implications that would be of value to investors, corporations, and policymakers in making better decisions.

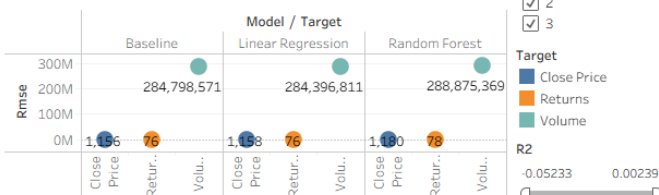
## Final Assignment

### Model Evaluation Summary

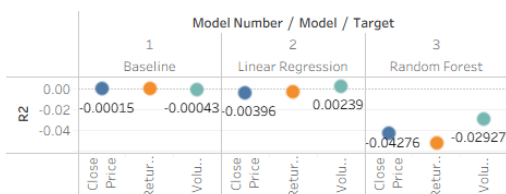
#### MAE Analysis by Model Type



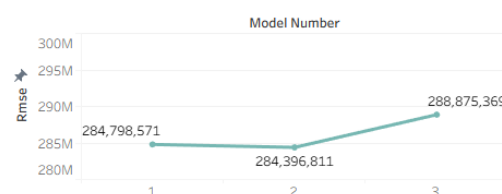
#### RMSE Analysis by Model Type



#### R-Square Analysis by Model Type



#### Model Error Benchmarking



### Model Limitations:

Even though the models created during this research are helpful regarding the assurance of connections between macroeconomic factors and the performance of the stock market, they have a number of limitations that influence their predictive power and efficiency in general.

Among the key shortcomings is the fact that macroeconomic variables are slow moving indicators. Variables like GDP growth, inflation and unemployment are usually updated monthly or quarterly but stock market data is updated on a daily basis or even a few seconds. This frequency mismatch renders the models hard to adopt any short-term market variations.

The other notable weakness is that the models lack investor sentiment and behavioural issues, which contribute significantly in financial markets. The news events, geopolitical events, and investor psychology are usually the drivers of the market movements, which are not reflected in the dataset. This means that the models are incapable of explaining abrupt fluctuations in the market.

The study uses also traditional statistical and machine learning models, including Linear Regression and Random Forest. Although these models can be useful in determining relationships, they might not encapsulate the complex nonlinear relationships, or time-dependent relationships that are non-linear, which evidently exist in financial markets.

Moreover, high-frequency and intraday data are not covered in the dataset, and hence, the models could not be used to determine the short-term volatility and patterns of trading. The lack of market microstructure information, e.g. order book data, or bid-ask spread, further limits the analysis.

The other drawback is associated with structural disruptions and crisis periods. Incidences like the 2008 financial crisis are known to change market behaviour and the models might be not entirely adjusted to changes in economic conditions which occur abruptly.

### **Key Limitations:**

- Indicators of macroeconomic movements were slow-paced in comparison to the swiftly floating market data.
- Lack of investor sentiment, news and behaviour factors.
- Weakness of models to embrace complicated nonlinear associations.
- Inadequacy of high-frequency financial data.
- Not all changes in market structure in periods of crisis.

### **Business Impact and Conclusion:**

#### **Business Impact**

The findings of this research are interesting in terms of their impact on the financial market behaviour and how the findings can be used to make a superior decision by investors, corporations, and policy makers. The knowledge of how the economic conditions are connected to the performance of the stock market will enable the stakeholders to have better interpretation of market trend and make strategic decisions.

To investors and managers of investment portfolios, the macroeconomic data on GDP growth, inflation, and interest rates could also be valuable indicators in modifying investment policies. Financial markets tend to be more profitable in terms of corporate earnings and better performance

of stocks when the economy grows and inflation is stable. In this scenario, investors can invest more with equity markets. Conversely, the occurrences of a growing inflation, rise in interest rates or the growth in unemployment might point to a slowdown in an economy whereby, investors are more likely to engage in defensive forms of investment as a way of minimizing financial risk.

The analysis of the trading volume is also helpful to the investors. The increased trading volumes tend to mean that the market is experiencing increased participation and liquidity whereas the decreased volumes tend to signal reduced investor confidence. Following these indicators, investors would be in a better position to understand the market environment and make decisions impulsively when the market is on a short-term fluctuation.

In the case of a corporation, the macroeconomic analysis assists in long term strategic planning. Economic growth normally presents firms with a possibility of expanding production, growing business, investing in research and development, and raising funds. Conversely, the period of economic decline forces companies to concentrate on cost management, efficiency, and risk management. It is through monitoring macroeconomic indicators like consumer demand, unemployment levels and interest rates that organizations can predict the changes in the economic environment and respond to the changes through adjusting their strategies.

The interaction between the macroeconomic conditions and financial markets can be also helpful to policymakers and financial regulators. Monitoring of these relations assists governments to determine economic risks, ensure financial stability and formulate effective economic policies. To illustrate, any alteration in interest rate policies or fiscal expenditures can have a strong impact on the investment behaviour or performance of the financial markets. The dynamics can be used to understand how policymakers can enact balanced policies that can foster sustainable economic growth.

On the whole, the analysis shows that macroeconomic indicators are helpful in terms of interpretation of the trends in financial markets and may be used to make more informed decisions in various fields.

### **Recommendations:**

Following the findings of this research, a number of suggestions can be offered concerning the way to improve financial decision-making and future studies.

- Investors are advised to include macroeconomic indicator in the long term investment plans as opposed to using a short term market trend.
- Corporations ought to be keen on macro-economic conditions when they seek to expand their business, plans of production, and investments.
- Policymakers must also take into account how the economic policy decision-making and the financial market reactions would interact when the fiscal and monetary policies are introduced.
- Future studies may involve more factors like investor mood, financial news coverage and industry specification indicators to enhance forecasting models.
- More sophisticated methods of analysis based on time-series forecasting models or deep learning would be considered to extract non-linear relationships in financial markets.

### **Conclusion:**

The current project explored the connection between the macroeconomic indicators and the performance of the stock market based on the dataset of stock market performance from 2000 to 2025. The study used the methods of exploratory data analysis, statistical modelling, and machine learning to examine the effect of economic variables on the financial markets in the form of GDP growth, inflation, interest rates, unemployment, corporate profits, and government debt.

These results indicate that macroeconomic variables are significant in prediction of long-term financial markets. The good economic growth is likely to increase the prices of the stock whereas increasing inflation and interest rates mostly create downturn on the performance of the market. These correlations are in line with the available financial theory that tend to associate financial conditions with the profitability of the corporation or expectations of investors.

Nevertheless, the predictive models designed within the framework of the given project also indicate the shortcomings of the financial market forecasting based solely on the macroeconomic variables. The returns of stock markets are very volatile and sensitive to various external factors such as the sentiments of investors, geopolitical events, and various economic shock events. Consequently, the predictive models are less effective when trying to predict the short-term market movement.

Nevertheless, the analysis gives credible informational bases to the structural relationship between the economic conditions and financial markets despite these limitations. The project will help to understand more about the responsiveness of financial systems to the changes in the economic conditions by finding the trends in the macroeconomic indicators and market behaviour.